CAREFLOW – CLINICAL PATHWAY PROCESS MINING

Mid-Review Report

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1. Project Overview

CareFlow – Clinical Pathway Process Mining is a healthcare process analytics project designed to understand and analyze patient journeys through hospital workflows.

The project uses timestamped hospital event logs to study the sequence of activities performed during patient journeys. Process mining and data analytics techniques are used to discover the actual workflow, identify repeated activities, detect potential bottlenecks, analyze waiting and transition times, and compare actual patient pathways with an expected workflow.

The overall project workflow is:

Hospital Event Data

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Data Preparation & Cleaning

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Exploratory Data Analysis

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Bottleneck & Waiting-Time Analysis

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PM4Py Process Discovery

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Process Model / Process Map

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Conformance Checking

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Power BI Dashboard

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Healthcare Process Insights

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2. Problem Statement

Hospital workflows involve multiple activities such as registration, triage, consultation, diagnostic tests, review, pharmacy, and discharge.

Traditional dashboards can provide numerical information such as patient counts and waiting times, but they may not clearly represent the actual sequence of activities followed by patients.

Patients may experience:

- Repeated activities
- Long waiting times
- Process loop-backs
- Different patient pathways
- Deviations from the expected workflow
- Inefficient transitions between activities

One of the observed patterns in the project is:

Triage
↓
X-Ray
↓
Triage

The repeated Triage activity represents a process deviation that requires further investigation.

CareFlow aims to use process mining and data analytics to identify, quantify, and visualize such process patterns.

3. Project Objectives

The main objectives of CareFlow are:

- Generate realistic timestamped hospital event logs.
- Clean and standardize the event data.
- Perform exploratory data analysis.
- Understand patient activity sequences.
- Identify repeated activities and potential bottlenecks.
- Analyze transition and waiting times.
- Discover the actual hospital workflow using PM4Py.
- Generate formal process models.
- Compare actual patient journeys with the expected workflow.
- Perform conformance checking.
- Build an interactive Power BI dashboard.
- Present meaningful process insights.

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4. Weekly Progress

Week 1 – Project Setup, Data Preparation & Analysis

- Set up and reviewed the CareFlow project structure and documentation.
- Updated the **README and project documentation** with the completed work and findings.
- Generated and cleaned the hospital event log using Python and Pandas.
- Performed EDA and analyzed patient activity sequences, repeated activities, and waiting times.
- Identified the **Triage → X-Ray → Triage** process loop involving **73 out of 200 patients (36.5%)**.

- Reviewed the project code and verified the major components of the workflow.

Week 2 – Process Mining, Dashboard & Mid-Review Preparation

- Tested the complete project from a **fresh clone** and verified that the event generation, cleaning, EDA, and PM4Py workflow were functioning correctly.
- Performed process discovery using **PM4Py**, generating the Directly-Follows Graph (DFG) and Petri Net process model.
- Completed conformance checking and obtained **63.5% conformant** and **36.5% deviating** cases.
- Continued development of the **Power BI dashboard**, including core KPIs and process-analysis visuals.
- Updated the **README and project documentation** with the latest results and progress.
- Prepared and organized the **Mid-Review document and PPT**, including project findings, process models, conformance results, and dashboard outputs.

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5. Data Preparation

5.1 Event Log Generation

A Python-based event generator was developed to simulate hospital patient journeys.

The event log contains:

- Unique patient identifiers (Case_ID)
- Hospital activities (Activity_Name)
- Activity timestamps (Timestamp)

The generated dataset contains:

- **200 patient cases**
- **873 events**

5.2 Data Cleaning

The clean_events.py script was used to prepare the raw event log for further analysis.

The cleaning process included:

- Missing value checking
- Timestamp validation
- Duplicate checking
- Data standardization
- Preparation of the cleaned event log

The cleaned dataset contains:

Original rows: 873

Final cleaned rows: 873

Rows removed: 0

Unique patients: 200

The cleaned event log is stored as:

data/hospital_event_log_clean.csv

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6. Exploratory Data Analysis

Exploratory Data Analysis was performed using Python and Pandas.

The analysis focused on:

- Dataset overview
- Number of patients
- Number of events
- Activity distribution
- Average events per patient

- Repeated activity analysis
- Triage → X-Ray → Triage loop analysis
- Waiting-time analysis
- Activity visualizations

The analysis identified a repeated Triage pattern involving:

73 out of 200 patients = 36.5%

This repeated pattern became the main focus of the subsequent bottleneck and process-mining analysis.

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7. Bottleneck Analysis

The bottleneck analysis focused on identifying repeated and potentially inefficient process patterns.

The main process pattern identified was:

Triage
↓
X-Ray
↓
Triage

Out of the 200 patient cases:

- **73 patients** experienced the Triage → X-Ray → Triage loop.
- This represents **36.5% of all patient cases**.

The repeated Triage pattern was therefore identified as the primary process deviation requiring further investigation.

The finding was initially identified through EDA and was subsequently examined using PM4Py process discovery and conformance checking.

Status: Completed

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8. Waiting-Time Analysis

Waiting-time analysis was performed to understand the time spent between consecutive activities in the patient journey.

Particular attention was given to transitions associated with the identified process loop, including:

X-Ray → Triage

The analysis also considered important transitions such as:

Triage → X-Ray

X-Ray → Triage

Triage → Doctor Consultation

Doctor Consultation → X-Ray

Waiting-time analysis provides additional information about process delays and helps complement the bottleneck analysis.

The resulting information is being incorporated into the Power BI dashboard through the **Average Wait Time** KPI and transition/waiting-time visuals.

Status: Completed / Dashboard Integration

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9. PM4Py Process Discovery

PM4Py was introduced to discover the actual patient workflow from the cleaned event log.

The cleaned event log was used as the input for process discovery.

The process-mining workflow successfully generated:

- Directly-Follows Graph (DFG)
- Petri Net process model
- Process-map visualization

The process discovery workflow was also tested successfully from a fresh clone of the project.

Status: Completed

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10. Directly-Follows Graph

The Directly-Follows Graph represents transitions between consecutive hospital activities.

It helps identify:

- Common activity transitions
- Frequently occurring process paths
- Process loops
- Repeated transitions
- Different patient activity sequences

The DFG provides a formal process-level representation of the patterns initially identified during EDA.

The following loop-back pattern was observed:

Triage

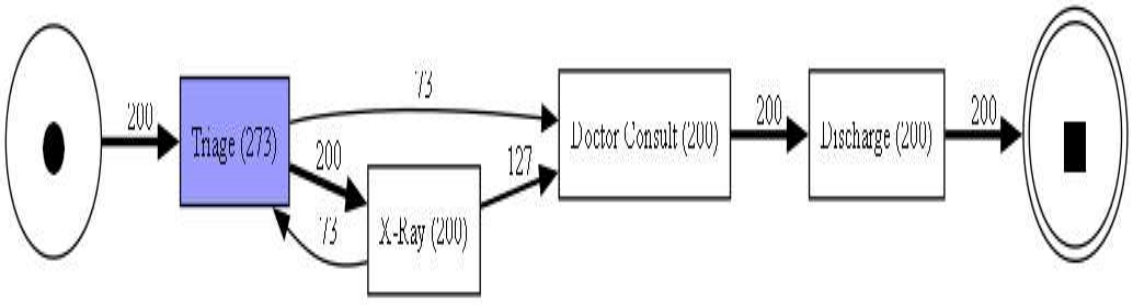
↓

X-Ray

↓

Triage

The DFG therefore provides visual evidence of the repeated process behavior in the event log.



11. Petri Net Process Model

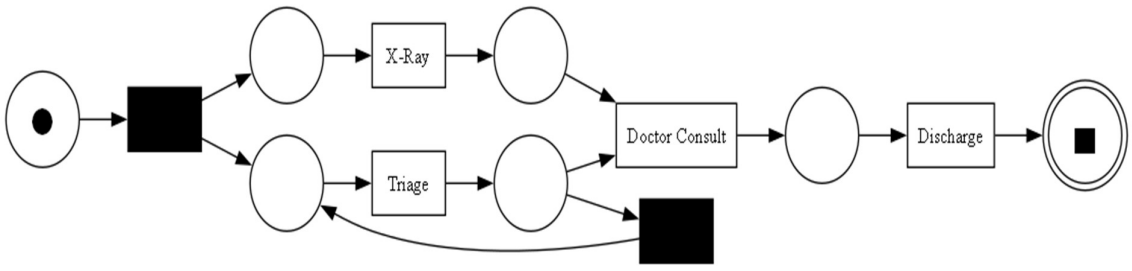
A Petri Net process model was generated using PM4Py.

The Petri Net provides a structured representation of:

- Activities
- Process transitions
- Possible patient paths
- Process flow relationships
- Repeated process behavior

The generated Petri Net model was successfully verified during project testing.

Status: Completed



12. Conformance Checking

Conformance checking was performed to compare the actual patient activity sequences against the expected hospital workflow.

The reference workflow used for the analysis was:

Triage

↓

X-Ray

↓

Doctor Consult

↓

Discharge

The **200 patient cases** were evaluated against this expected pathway.

12.1 Conformance Results

Metric	**Result**
Total Patients	200
Conformant Patients	127
Deviating Patients	73
Conformance Percentage	63.5%
Deviation Percentage	36.5%
Average Wait Time	13.10

12.2 Key Finding

Out of 200 patient cases:

- **127 patients (63.5%)** followed the defined expected pathway.
- **73 patients (36.5%)** deviated from the expected pathway.

The **36.5% deviation rate** is consistent with the repeated Triage pattern identified during earlier analysis.

The same process pattern was observed across multiple stages:

1. EDA identified the Triage → X-Ray → Triage loop.
2. PM4Py process discovery represented the loop in the discovered process.
3. Conformance checking identified 73 deviating cases.

This consistency provides stronger evidence that the repeated Triage pattern represents a significant process deviation in the generated event log.

Status: Completed and Verified

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13. Power BI Dashboard

Power BI is being used to consolidate the process-mining and healthcare operational findings into an interactive dashboard.

The dashboard focuses on presenting the key process KPIs and supporting process-analysis visuals.

13.1 Core KPI Cards

The dashboard includes four main KPI areas:

Total Cases

Displays the total number of patient cases analyzed.

Value: 200

Conformant %

Displays the percentage of patient cases that followed the defined expected pathway.

Value: 63.5%

Deviating %

Displays the percentage of patient cases that deviated from the expected pathway.

Value: 36.5%

Average Wait Time

Displays the average waiting time between relevant patient activities.

Value:13.10

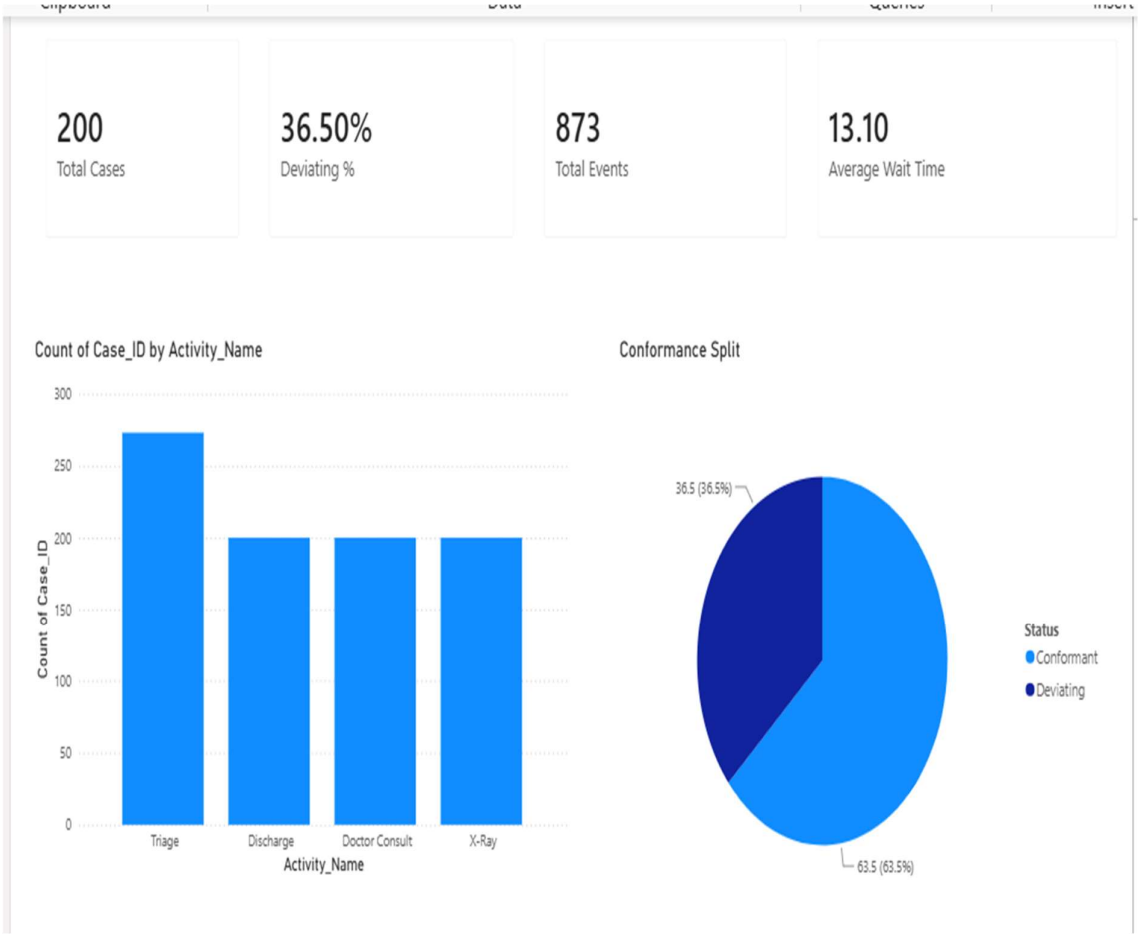
13.2 Dashboard Analysis

The dashboard includes or is being developed with:

- Total Cases KPI

- Conformant % KPI
- Deviating % KPI
- Average Wait Time KPI
- Activity counts
- Conformance split

The dashboard allows the project findings to be viewed from both a high-level KPI perspective and a detailed process perspective.



14. Technology Stack

Technology	Purpose
Python	Event generation and data processing
Pandas	Data cleaning and analysis
Matplotlib	Data visualization
CSV	Event-log storage
PM4Py	Process mining and discovery
Graphviz	Process visualization
Git	Version control
GitHub	Project collaboration
Power BI	Interactive dashboard

Deferred Technologies

Technology	Purpose	Status
BigQuery	Data warehousing	Deferred
dbt	Data transformation	Deferred

BigQuery and dbt were evaluated but deferred from the current implementation so that the project could remain focused on the core process-mining and dashboard workflow.

15. Fresh-Clone Testing

The complete project was tested from a fresh clone to verify that the implementation works correctly in a clean environment.

The following components were checked:

- Repository cloning
- Dependency installation
- Event generation
- Data cleaning
- EDA workflow
- PM4Py process discovery
- DFG generation
- Petri Net generation
- Process-map generation
- README setup instructions

The PM4Py process discovery workflow successfully executed and generated the expected process-mining outputs.

Fresh-Clone Testing Status: PASS

This verification helped ensure that the project is reproducible and that the documented setup instructions are consistent with the actual implementation.

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16. Key Findings

The major findings obtained so far are:

Finding 1 – Repeated Triage Pattern

A repeated Triage pattern was identified in:

73 out of 200 patients = 36.5%

Finding 2 – Process Loop

The major repeated process pattern is:

Triage → X-Ray → Triage

Finding 3 – Conformance

The expected workflow was defined as:

Triage → X-Ray → Doctor Consult → Discharge

The conformance analysis resulted in:

- **127 conformant cases – 63.5%**
- **73 deviating cases – 36.5%**

Finding 4 – Process Discovery

PM4Py successfully generated both the DFG and Petri Net process model.

Finding 5 – Dashboard

The Power BI dashboard consolidates the major findings through KPI cards and process-analysis visuals.

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17. Current Project Status

Component	Status
Event generation	Completed
Raw CSV generation	Completed
Data cleaning	Completed
Data dictionary	Completed
EDA	Completed
Repeated Triage analysis	Completed

Component	Status
Bottleneck analysis	Completed
Waiting-time analysis	Completed
PM4Py process discovery	Completed
DFG generation	Completed
Petri Net generation	Completed
Process-map analysis	Completed
Fresh-clone testing	Completed
Conformance checking	Completed
Conformance results validation	Completed
Power BI core KPIs	Completed
Power BI dashboard refinement	Completed
README updates	Completed
Mid-Review document	Completed
Mid-Review PPT	Completed
BigQuery	Deferred
dbt	Deferred

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18. Mid-Review Deliverables

The Mid-Review will demonstrate the progress achieved across data preparation, process analysis, process mining, conformance checking, and dashboard development.

Data Preparation

Data Generation

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Data Cleaning

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EDA

Process Analysis

Repeated Activity Analysis

↓

Bottleneck Identification

↓

Waiting-Time Analysis

Process Mining

PM4Py Process Discovery

↓

Directly-Follows Graph

↓

Petri Net

↓

Process Interpretation

Conformance & Dashboard

Conformance Checking

↓

Power BI Dashboard

↓

Process Insights

The Mid-Review presentation includes:

- Introduction
- Problem statement & Objectives
- Project Workflow
- Technology stack
- Data preparation & cleaning
- EDA & initial findings
- Bottleneck analysis & Waiting-time analysis
- PM4Py process discovery
- PM4PY outputs Directly-Follows Graph & Petri Net process model
- Conformance-checking
- Conformance findings & results
- Power BI dashboard
- Conclusion

20. Conclusion

CareFlow has progressed from hospital event-log generation and data preparation to exploratory analysis, bottleneck identification, waiting-time analysis, PM4Py process discovery, conformance checking, and Power BI dashboard development.

The main process pattern identified is:

Triage → X-Ray → Triage

This pattern affects **73 out of 200 patient cases (36.5%)**.

Conformance checking against the expected workflow:

Triage → X-Ray → Doctor Consult → Discharge

identified **127 conformant cases (63.5%)** and **73 deviating cases (36.5%)**.

The consistency between the EDA findings, PM4Py process discovery, and conformance results provides a clear indication of the observed process deviation.

The Power BI dashboard is being developed to consolidate these findings using core KPIs such as **Total Cases, Conformant %, Deviating %, and Average Wait Time**, along with supporting process and transition analysis.

In addition to the technical implementation, project reproducibility was verified through fresh-clone testing, and the README and project documentation were continuously updated to reflect the latest progress and results.

The current focus is on finalizing the Power BI dashboard, completing the Mid-Review documentation and presentation, validating all reported metrics, and preparing the project demonstration.